Pearson
BTEC

Mark Scheme (Results)

January 2024

Pearson BTEC Nationals in
Applied Science (31617H/1C)
Unit 1: Principles and Applications of Science I
Chemistry

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Unit 1: Principles and Applications of Science I Chemistry

General marking guidance

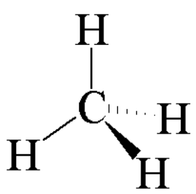
- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

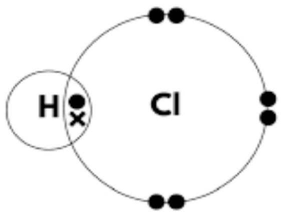
Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark
1(a)(i)	**A** 6 6		1
1(a)(ii)	$1s^22s^22p^2$	do not accept subscript 2 do not accept electron in box diagram accept uppercase p reject any additional orbitals	1
1(b)(i)	fire extinguisher / dry ice / coolant / fizzy drinks allow any other correct use	accept dry cleaning accept carbonated drinks accept (to promote) **photosynthesis**	1
1(b)(ii)		accept single lines for each bond accept dotted and full bond can be in either position accept h for H reject molecules with all 4 bond angles at 90º reject any number of hydrogens other than 4	1
1(b)(iii)	an explanation linking C=O has a **double bond** (1) C=O has two shared pairs of electrons (1) strong**er** attraction between (shared pairs of) electrons and nuclei of atom (1) so therefore more **energy** to break (the bond) (1)	accept ORA throughout ignore C=O alone as given in stem must refer to a double bond ignore references to bond length	4
			Total = 8 marks

Question Number	Answer	Additional Guidance	Mark
2(a)(i)	**C** chlorine		1
2(a)(ii)	an explanation linking any **three** from boiling points increase (down the group / table) (1) larg**er** atoms / more electrons (1) {larg**er** / strong**er**} van der Waals forces (1) more {**heat** /**energy**} needed to {separate the molecules / break the (intermolecular) forces}(1)	ignore references to reactivity accept larg**er** molecules / more shells (of electrons) ignore more outer shells of electrons ignore more shielding accept {larg**er** / strong**er**} {London dispersion forces / intermolecular forces} accept more {**heat** /**energy**} to break the **intermolecular** bonds	3
2(b)(i)	 (2) or **one** shared pair of electrons between H and Cl (1) rest of molecule correct (1)	accept dots, crosses or a mixture of both accept no circles ignore any inner shell electrons, correct or otherwise	2
2(b)(ii)	an explanation linking (large) difference in electronegativity / {electron pair is / electrons are} more attracted to the Cl (1) so H is ∂+ **and** Cl ∂– (1)	accept annotated diagram for both marks accept uneven distribution of electrons accept H is positive **and** Cl is negative ignore references to intermolecular forces reject references to ionic bonds for both marks	2
			Total = 8 marks

Question Number	Answer	Additional Guidance	Mark
3(a)	**A** high high		1
3(b)	Fe = 0 Fe_2O_3 = +3	accept zero accept 3 / three / 3+ accept superscript with symbol reject -3 for MP2	2
3(c)(i)	<u>calculation of number of moles Fe (1)</u> 27.90 ÷ 55.8_ = (0.5) <u>calculation of mass of Fe_2O_3 (1)</u> (0.5) ÷ 2 x 159.6 <u>evaluation (1)</u> 39.9 g allow alternative methods	39.9 gains three marks with or without working accept ECF accept correct rounding	
3(c)(ii)	<u>substitution (1)</u> 19.53 ÷ 27.90 x 100 <u>evaluation (1)</u> 70 (1)	70 gains two marks with or without working accept ECF	2
			Total = 8 marks

Question number	Indicative content
4	<u>similarities</u> • all turn universal indicator purple because they all produce a hydroxide • which is alkaline / has a high pH • all float because the density is less than water • all fizz because they produce hydrogen gas • all move on the top of the water because they all produce hydrogen • metal + water → metal hydroxide + hydrogen • $2Li + 2H_2O \rightarrow 2LiOH + H_2$ • $2Na + 2H_2O \rightarrow 2NaOH + H_2$ • $2K + 2H_2O \rightarrow 2KOH + H_2$ • all react in same way because they are all in group 1 • all have one electron in outer shell/orbital <u>differences</u> • all make different hydroxides because a different metal is reacted • potassium moves most quickly because it is the most reactive • only potassium produces a flame as it is most reactive • the metals get more reactive as you move down the group • the further down the group the more shells of electrons the atoms have • potassium has the largest atomic radius • outermost electron is furthest from nucleus • more shielding as you move down the group • weakest nuclear attraction • easier to lose an electron as you move down the group • energy needed to form positive ions decreases • potassium is the only one that pops because it gives out the most heat energy / most exothermic • so lights the hydrogen

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes.

Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Adequate interpretation, analysis and/or evaluation of the scientific information with generalised comments being made. • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. • The explanation shows some structure and coherence.
Level 2	3–4	• Good analysis, interpretation and/or evaluation of the scientific information. • Lines of argument mostly supported through the application of relevant evidence. • The explanation shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Comprehensive analysis, interpretation and/or evaluation of all pieces of scientific information. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The explanation shows a well-developed structure which is clear, coherent and logical.

Llywodraeth Cynulliad Cymru
Welsh Assembly Government

Rewarding Learning